Journal of Big Data

RESEARCH Open Access

An effective approach for classification of plant disease of small size dataset by using sequential deep learning model with Min–Max Hue histogram based segmentation

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Abstract

Training deep learning models for plant disease classification on small size datasets with heterogeneous backgrounds remains challenging issue. Proposed study overcomes the major limitations and weaknesses of thexisting contemporary methods. This study introduces a novel three-phase approach: (1) segmentation of disease regions using a Min–Max Hue Histogram-based technique to isolate the disease region with the most important data, (2) a lightweight, sequential deep learning model (PDC-Net) trained from scratch to classify the diseases and (3) a disease recovery module to provide recommendation to farmer. Unlikexisting works, we avoid data augmentation and transfer learning toliminate overfitting risks and domain mismatches. Additionally, we created the Plant Disease Small Dataset (PDSD)—a new, realistic subset of the PlantVillage dataset featuring images with variednvironmental conditions and backgrounds.experimental results show that our method achieves an average classification accuracy of 91%, which is better than popular CNN architectures (VGG19 and ResNet50). This complete system provides an accurate,ffective, and convenient way for farmers to diagnose plant diseases and to receive recovery strategy, which can help promote the precision agriculture practices.

Keywords: Contrast stretching, RoF filter, GIF filter, Hue histogram, HSI color model, K-mean clustering, Segmentation

Introduction

To maximize the crop yield, Indian farmers try to protect their crops from disease by regularly checking the crops and spotting disease at anarly stage [1]. However, as simple as it may sound, the detection of plant diseases is not a simple task. Identification of these diseases requires specialized knowledge, which farmers usually lack. As a result, the farmers would need to consult specialists who can help them to identify plant illnesses, which may be fairlyxpensive, time-consuming, and the results are frequently unreliable. For the reasons outlined above,xtensive study has been carried out to develop techniques that will be sufficient in accuracy and available to the majority of farmers without consulting specialists.

Various studies have been carried out in the context of automatic identification of plant diseases using a variety of techniques, including image processing, machine learning (ML), deep learning (DL),tc. Deep learning, particularly convolutional neural networks (CNN), provides more accurate categorization and diagnosis of plant disease results than machine learning, and allowing us to make better decisions.

Although the aforementioned studies have made a substantial contribution to the field of automatic disease diagnosis, only a small number of them have addressed the problem of data scarcity, which has an impact on theffectiveness of DL-based architectures. To address the challenge of small-sized datasets, current deep learning techniques for plant disease diagnosis often rely on data augmentation or transfer learning. However, these methods present limitations, such as the risk of overfitting due to overly similar augmented images or the potential for negative transferffect in transfer learning [2]. Furthermore, mostxisting models inadequately address the critical need for precise disease spot segmentation, which isssential for accurate classification.

This paper addresses the practical challenges inherent in current precision farming strategies by proposing a novel approach designed to overcome identified limitations andmpower farmers withnhanced decision-making capabilities for improved crop health. To address these critical research gaps, this study makes the following major contributions:

- We introduce a novel three-phase framework that includes (1) Min–Max Hue Histogram-based segmentation, (2) a custom sequential deep learning model (PDCNet) trained from scratch, and (3) a disease recovery recommendation module.
- We created a new subset of the PlantVillage dataset called the Plant Disease Small Dataset (PDSD), focusing on images with complex backgrounds and diversenvironmental conditions, which better simulate real-world scenarios.
- Our approach demonstrates superior performance over state-of-the-art methods, particularly for small size datasets, by segmenting only the relevant disease regions and ignoring irrelevant background information.

Thus, this work fills important gaps in the identification of plant diseases on a small scale and provides a reliable, affordable, and useful recommendation system for precision farming.

The rest of the paper is structured as follows: Sect. "Related works" deals with recent work done in the field of disease classification with deep learning. Sect. "CNN-based disease detection" discusses a proposed working model of PDCNet (Plant Disease Convolutional Network) for detecting leaf disease using deep neural networks. Sect. "Hybrid andnsemble models" deals withxperimental results and discussion. Summary and remarks are described in Sect. "Pretrained and transfer learning approaches".

Related works

Automatic classification of plant diseases has advanced significantly in recent years, largely driven by the adoption of deep learning techniques. This literature review provides a comprehensive overview of past studies, highlighting their favorable results and persistent challenges.

CNN-based disease detection

Initial studies, such as Mohantyt al. (2016), pioneered the use of CNN architectures like AlexNet and GoogleNet on the PlantVillage dataset, achieving a remarkable 99.35% accuracy across 38 plant disease classes [3]. Building on this, Toot al. (2019) advanced the work by utilizing Inception-V3 for disease-specific classification, including bacterial spot and healthy leaves, with their model reaching an impressive 99.76% accuracy [4]. Karlekarat al. (2020) presented a two-stage architecture, SoyNet, which combined CNN classification modules with image processing modules (IPM). Their methodffectively isolated leaf regions by subtracting complex backgrounds, leading to improved classification accuracy [5]. Kumart al. (2021) further reinforced thefficacy of CNNs for plant disease detection across varied agricultural conditions [6].

However, challenges remain. So, Panchalt al. (2021) proposed a CNN-based model, trained on a publicly available dataset of 87,000 RGB images, which achieved 93.5% recognition accuracy. Despite this, their model suffered from class misclassification due to dataset limitations in size and diversity, resulting in decreased performance in real-world applications [7]. Similarly,nkvetchakult al. (2021) introduced a loss-fused, resilient CNN model tested on the PlantVillage benchmark dataset. While it attained 98.93% accuracy in controlled settings, it failed to maintain performance under diverse real-world conditions [8]. Pradeept al. (2022) suggested a CNN-basedfficientNet model for multi-label and multi-class plant disease classification, utilizing deep featurextraction via hidden layers; however, its performance on benchmark datasets was not sufficient [9]. More recently, Tashin Ahmedt al. (2023) introduced a two-phase CNN architecture for rice crop disease detection. Their model, utilizing a limited dataset of only 200 images across three classes,fficiently identified regions of interest in the initial phase and classified them in the subsequent phase, demonstrating promising results despite the small dataset size [10].

Despite significant progress in this area, several persistent challenges need to be addressed. Manyxisting datasets are collected under controlled conditions, failing to reflect real-world variability, which often leads to models struggling with generalization in diverse field conditions. Moreover, these models frequently demand substantial computational power, and their results can suffer from overfitting or inadequate generalization to actual field conditions.

Hybrid andnsemble models

A CNN-SVM hybrid model was presented by Narayanant al. (2022) for the categorization of banana leaves. The method used a CNN, and a fusion support vector machine (SVM) based classification system. Their approach utilized median filtering for preprocessing. Their accuracy was 99 However, the methodology lacked classification diversity [11]. Astanit al (2022) proposednsemble classifiers whichvaluated using the PlantVillage and Taiwan Tomato Leaves Datasets. Theirxperiments demostrated promising results [12].

Pretrained and transfer learning approaches

Chent al. (2020) used a transfer learning technique with ResNet50 to classify plant diesases. Their result achieved an accuracy rate of 98.9% [13]. Jadhavt al. (2020) proposed transfer learning-based plant disease detection system which utilized AlexNet and GoogleNet for classification [14]. In addition to these, Kabirt al (2020) proposed a multi-label CNN model for classifying various plant diseases using transfer learning techniques [15], including AlexNet [16], GoogleNet [17], VGG19 [18], ResNet50 [19], ImageNet [20], Dense121 [21], LeNet [22], and XceptionNet [23]. The authors claimed theirs was the first study to use a multi-label CNN to classify 28 different plant disease subtypes. Althoughffective in specific contexts, these models showed reduced performance when applied to more diverse, real-world datasets.nkvetchakul and Surinta (2022) introduced a CNN-based methodology that used transfer learning to categorize two particular plant diseases. The NASMobileNet and MobileNetV2 networks weremployed in their model. In theirxperiments, NASMobileNet gave them more accurate predictions. A number of different data augmentation procedures, including rotation, zooming, brightness alteration, and mixing, were used by them in order to mitigate the issue of overfitting. They achieved a maximal test accuracy of 84.51% by utilizing the Cassava 2019 a separately collected leaf disease dataset [24].

To address the issue of data scarcity, Arsenovict al. (2019)xpanded their research by creating a large dataset of plant images captured under varying weather conditions, angles, and daylightxposures. They also proposed a GAN-based augmentation method combined with a two-stage detection algorithm tailored for real-timenvironments [25]. In addition, Abbast al. (2021) and Anht al. (2021)mployed pre-trained CNNs like MobileNet and generated synthetic data using GANs [26, 27].

Even though many researchers use data augmentation, it often creates duplicate images, which can lead to overfitting (Cogswellt al., 2015). Negative transfer is also barrier for transfer learning when the source and destination domains are significantly different [28].

Recent research has advanced the field of plant disease detection by paying more attention to small datasets, lightweight models, and sophisticated learning techniques. Forxample, PlantXViT (2022) achieved more than 93% accuracy by combining CNN and Vision Transformer (ViT) components together. With just 0.8 M parameters, they achieved good results [29]. MSNet (2023) combinedfficientNet and DenseNet with attention modules, in order to achieve good results while using very few resources [30]. Moreover, transformer topologies can achieve high accuracyven with minimal training data, as demonstrated by Zhaot al. (2022), who presented a novel transformer-based model for the purpose of plant disease detection [31]. While the newly introduced CNN network architecture with attention mechanisms by Ahmedt al. (2023) was able to reach a high level of accuracy in the identification of tomato diseases in authentic field images [32], Yangt al. (2024) looked into lightweight CNN architectures which were designed to operatefficiently on low-power peripheral devices, thereby addressing the challenge of deploying plant disease classifiers in resource-limitednvironments [33]. Lastly, Wangt al. (2024) proposed a self-supervised contrastive learning method which greatly improves the classification performance on small, mixed plant disease datasets [34].

These recent advancements indicate a growing trend toward utilizing compact andfficient models and self-supervised learning to address the challenge of limited data in plant disease detection. This direction aligns well with our approach, which integrates ROI-based segmentation with a shallow deep learning model to achievefficient and accurate solutions.

The PDCNet model mitigates the prevalent limitations observed inarlier plant disease classification models through the following key advancements:

- Reduces digital noise,nhancing visualization of disease spots.
- Does not require data augmentation or transfer learning.
- Identify disease areas prior to segmentation utilizing the hue histogram method.
- More precise segmentation with the help of Min–Max Hue Histogram and K-means clustering
- No GPU is needed for the proposed lightweight architecture.
- Achieved 91.25% accuracy, demonstrating superior performance on small, heterogeneous datasets.
- Developed a recovery recommendation system for farmers.

Proposed methodology

Tonhance learningfficiency from a limited collection of plant images, particularly those with significant background heterogeneity, and to provide the farmers with an actionable solution, the proposed system is logically divided into the following modules:

1.Leaf Segmentation Module (LSM).
2.Deep Learning Module (DLM).
3.Disease Recovery Module (DRM).

Figure 1 presents a comprehensive perspective of the proposed system. It demonstrated the proposed disease recognition and recovery procedures in a step-by-step manner:

If the CNN architecture is merely trained to locate the significant part only, it may perform better resultsven with a small size dataset. As a result, the first module works with segmenting plant images in order toxtract the major disease regions from the image. The original image is taken, resized to a predetermined size, and then sent to the pre-processing step, where Rank Order Fuzzy (RoF) filter and GIF (Guided Image Filtering) filter techniques are performed to remove undesirable noise and boost image contrast, respectively. On these improved images, we apply our proposed Min–Max Hue Histogram-based disease spot detection technique, in which image pixels with hue values between the min and max values of the hue histogram are combined with some threshold value to separate the disease regions from the leaves. Following the identification of important disease regions from an image, these regions arextracted using k-mean clustering algorithms. These images appear simply due to the absence of a heterogeneous background. Following that, these simplified images are fed into the proposed PDCNet architecture, which is trained with this simplified dataset to recognize disease.

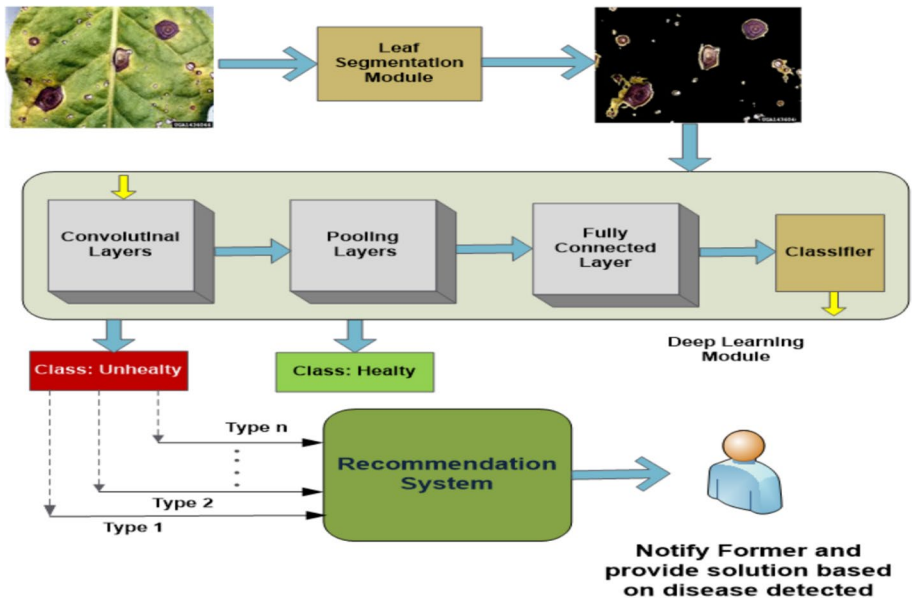

Fig. 1 Proposed working model

xperiments have demonstrated that the learning process is simple andffective. Finally, the disease recovery phase is carried out, where system can advise farmers on the disease and how to recover from it.

Detailedxplanations ofach module are provided in the following subsections:

Leaf segmentation module

In almost all of the images in the plant village database [35], it has been found that a portion of the leaf's surface is significantly lower than the portion of the background, and the amount of diseased tissue that can be found on a leaf isxtremely negligible in comparison to the whole background area. Therefore, the classification accuracy may be affected if CNN is trained without background removal. Alternatively, we can say that, before training the CNN, if the leaf disease portions of the image are separated from thentire image, it will be quite simple for the CNN to find and learn important features,ven if there are only a limited number of images available for training. Therefore, it isssential to perform background subtraction before training while the infected areas are still small. Here, we have applied LSM to completely remove the background from the image. LSM is made up of four sub-modules, and their results are shown in Fig. 2.

Image acquisition and preparation

In image acquisition, a leaf image is obtained and serves as the designated image for subsequent processing. In order to acquire and prepare image data, we used a portion of the PlantVillage dataset [35] that included 12 different types of healthy and infected leaves. This dataset comprises 3598 images of plant leaves with their respective labels. It includes a collection of images taken in different conditions.

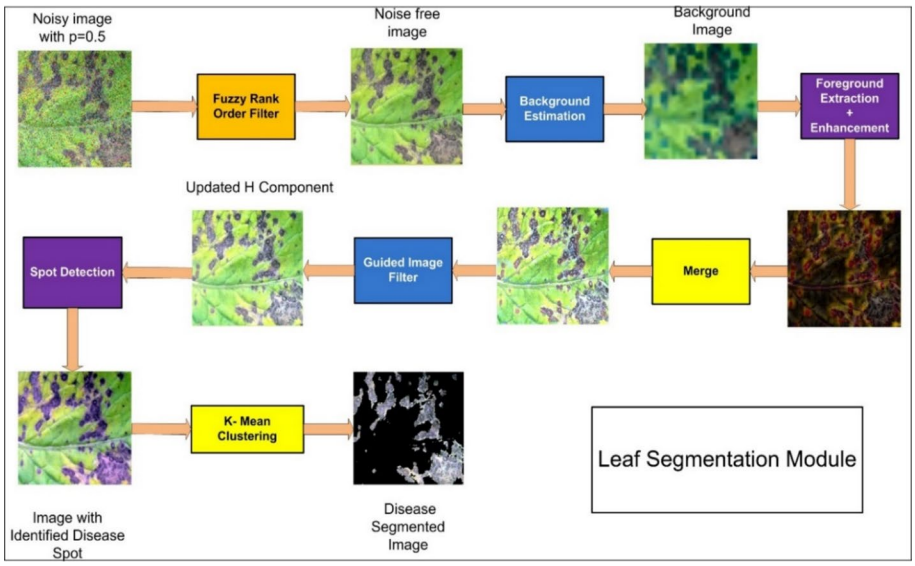

Fig. 2 Working of Leaf Segmentation Module

Pre-processing

Pre-processing of the input image is anssential step to improve the quality of the image andliminate undesired distortions. The pre-processing step consists of a combination of image denoising and foregroundnhancement methods that produce a higher-quality image. For image denoising, the RoF filter [36, 37] is used. Image improvement with the RoF filter is accomplished by the use of fuzzification, which transfers images from the spatial domain to the fuzzy domain,nhancement of fuzzy membership values, and defuzzification, which transforms the original image back to the spatial domain. The RoF filter is used tonhance the quality ofxtremely noisy images, particularly in cases where the noise level is highly distorted. RoF uses the ranking/density of noisy images as a fundamental step for noise detection. Next, toxtract foreground objects from the denoised image, we suggested a novel background subtraction method thatmploys a local pixel homogeneity scheme from CIE $L^* a^* b^*$ colour space [38] tostimate the background and a local intensity pattern comparison from the RGB colour model to detect the foreground. After detecting the foreground objects, these objects arenhanced by using an adaptive image improvement method called the GIF filter [39]. Based on the distribution features of the light intensity in a crop image, the GIF approach uses Otsu's thresholding [40] to automatically generatenhancement weights and adaptively change its contrast.

Leaf disease spot identification

Leaf disease spot identification is the third step of the leaf segmentation module. In this step, improved leaf images are processed by using the min–max hue histogram-based technique to identify an interested image area. Initially, the RGB picture undergoes a transformation to become an HSI color image. The H component adjustment is performed to provide a threshold value that distinguishes disease spots from plant

leaves. In order to do this, calculate the histogram of the hue component and identify the greatest and lowest peak points between 0 and 60°, in the histogram that correlate to the color associated with the illness [41]. These two points, designated $\tau 1$ and $\tau 2$, are used as thresholds.

Next, we use the following formula to find and isolate the illness spot inside the picture of the leaf.

$$H_{ij}{}' = \begin{cases} H_{ij} & : H_{ij} < \tau 1 \\ H_{ij} + \eta & : \tau 1 \leq Hij < \tau 2 \\ H_{ij} & : H_{ij} \geq \tau 2 \end{cases} \quad (1)$$

where the hue values of the input and output image are represented as H_{ij} and H'_{ij} respectively.η is another threshold that amplifies the hue value in the image in order to distinguish the diseases.

Following that, contrast stretching [42] technique is applied to the I component, which then results in an increase in the intensity factor globally.

Finally, combine the modified H and I components with the unchanged S component to get the final improved HSI color picture. This final improved HSI color map is then, converted back into the RGB color map that shows the image with illness spot identification [43].

Leaf segmentation

In the leaf segmentation step, the image is partitioned into various clusters by using K-mean clustering [44]. The K-mean clustering is applied on the L*a*b* color model [38]. After applying, the K-means clustering method, all of the clusters are then translated from the RGB color model to the HSI color model. A cluster is considered to be a disease cluster if it has a bigger hue mean value than the other clusters.

After getting the important part of an image, the smoothing filter is used to make the image look better. Because of the lack of a varied background, these images would appear simplistic.

Deep learning module

Pre-processing and segmentation techniques are used to separate the important area from the plant leaf that is the region of interest. The results from the LSM module are then put into classes so that identification and decisions can be made. The method known as Plant Disease Convolutional NETwork (PDCNet) is used for the classification process. In this step, the relevant information is obtained to classify the various crop disease types.

One of the branches of machine learning is known as "deep learning." Because it has numerous layers, a deep convolutional neural network (DCNN) is able to gradual-lyxtract higher level information from the images given as input. VGG19, ResNet50, GoogLeNet,tc. are some of the most widely used models for classifying images. These models perform poorly on datasets that are too specific. These networks can learn image information wellnough to classify images into a wide range of categories. When the differences between the visuals are not particularly noticeable, they struggle to learn.

In order to categorize plant diseases using limited datasets, the suggested network, which is referred to as PDCNet, was developed specifically for this purpose. Due to the fact that it is able to learn the more complicated properties of the dataset, it is able to accurately identify illnesses. The structure is defined in such a manner thatach following layer is able toxtract progressively complicated properties from the leaf data. The spatial resolution of a image is decreased withach successive layer, which has made it possible for us to see fine details of the leaf.

This section describes in detail description of the structure of PDCNet, which consists of the convolutional layer, activation functions that is rectified linear unit (ReLU) and softmax, the pooling layer, and the dropout layer. The subsequent subsections will present the detailed setup ofach layer that makes up PDCNet.

Model architecture

Figure 3 displays a visual illustration of the proposed PDCNet model. The PDCNet primarily consists of two fully connected layers F1 and F2, one flatten layer, five max-pooling layers (P1, P2, P3, ..., and P5), and five convolutional layers (C1,C2,C3,..., and C5). The convolutional layers are primarily responsible for improving the model's representational ability. The maxpool layer has 2×2 filters, whileach convolution network has a 11×11 filter. The architectures of PDCNet to support the different defined classes shown in Table 1.

Convolution layers

Convolution, which is the backbone of CNNs, is characterized as an operation on two functions. CNNs convolute the feature map with multiple inputs and pass the result to an activation function. Convolution in CNNs isxpressed as follows:

$$C_i = \Psi(I * Z_i) \tag{2}$$

The convolution operation, denoted as $*$, is applied to thextracted leaf image, I, using a kernel or mask Z_i in the ith layer. The resulting convoluted leaf image is then sent through the activation function $\Psi(.)$. The PDCNet consists of five convolutional

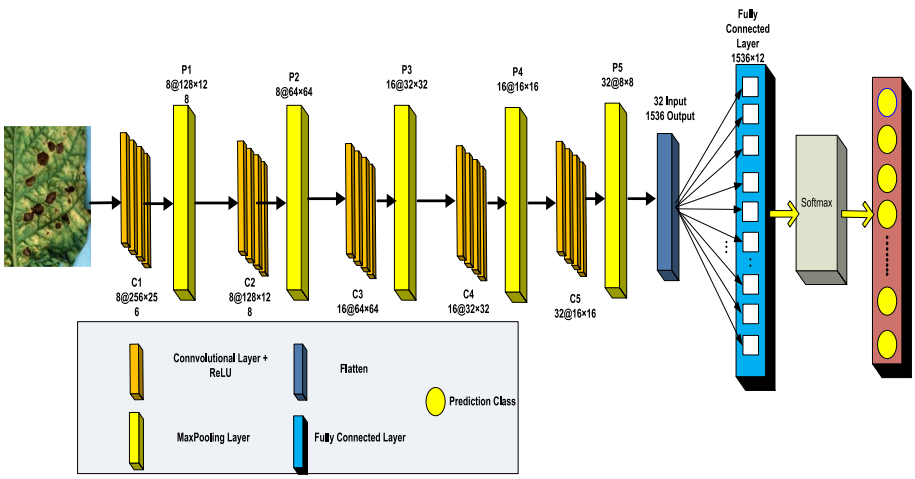

Fig. 3 Architecture of PDCNet

Table 1 Layars and its description of proposed pdcnet model

Function Name	Layer	Filter Size	Stride	Output tensor
Convolutional + ReLU	Conv1	11 ×11×8	1	8 ×256×256
Max Pooling	Pool1	2 ×2	2	8 ×128×128
Convolutional + ReLU	Conv2	11×11×8	1	8 ×128×128
Max Pooling	Pool2	2 ×2	2	8 ×64×64
Convolutional + ReLU	Conv3	11 ×11×16	1	16×64×64
Max Pooling	Pool3	2 ×2	2	16×32×32
Convolutional + ReLU	Conv4	11×11×16	1	16×32×32
Max Pooling	Pool4	2 ×2	2	16×16×16
Convolutional + ReLU	Conv5	11 ×11×32	1	32×16×16
Max Pooling	Pool5	2 ×2	2	32×8×8
Dropout	flatten	--	-	1536 ×1×1
SoftMax Regression	Output	--	-	12 ×1×1

layers,achquipped with an 11×11 filter. While there are only 8 neurons in the first two convolution layers, there are 16 neurons in the following two layers. The final convolution layers include 32 neurons.

Activation function

Rectified linear unit During the training phase, ReLU activation is chosen in all convolution network layers due to its superior performance in training over other activation functions like sigmoid and tanh. ReLU produces x as an output if the input, x, is positive;lse, it produces 0. The following is a mathematical definition of the ReLU function:

$$\mathrm{Re}LU(\upsilon) = \begin{cases} \upsilon & if\, \upsilon \geq 0 \\ 0 & otherwise \end{cases} \tag{3}$$

It is often referred to as the ramp function. It requires the input C_i, which is derived by applyingq. (2).

Softmax The softmax activation is used in the final dense layer to calculate the probability of predicted classes. The output is chosen to be the class with the highest probability. The formula for the softmax function is:

$$S(y_i) = \frac{e^{y_i}}{\sum_{j=1}^{k} e^{y_j}} \tag{4}$$

In this work, the variables e^{y_i} and e^{y_j} represents the likelihood of belonging to the categories i and j respectively. The variable K represents the number of categories and it is initially set to twelve in this work. The prediction forach class is determined using the softmax function $S(y_i)$.

Pooling layer

Pooling layer is responsible for lowering the overall output of the convolutional layer, and, as a result, lowering the number of trainable parameters. When a large number of convolutional layers are used, the training parameters also grow so quickly. Pooling layers are used to solve this issue. The pooling layer uses a region's overall spatial characteristic to represent thentire region, result in reducing the convolutional layer's output and trainable parameters. The PDCNet uses max pooling with a 2×2 window size. The maximum value that is present in the 2×2 window is what the max pooling operation reflects.

Loss function

The loss function is responsible for measuring the disparity between the output that was anticipated and the output that was wanted (the label value). PDCNet uses categorical cross-entropy to calculate loss. The following is the categorical cross-entropy function:

$$L = -\sum_{j=1}^{k} \frac{y_j \log(\hat{y}_j)}{K} \tag{5}$$

where y and $\hat{y}$ represent the anticipated and the desired outputs respectively.

Optimization and weight updation

The primary objective in the process of training a neural network is to determine the optimal value of a parameter, denoted as θ, that minimizes the loss function L. In order to facilitatexpedited convergence during training, the model utilizes the Adaptive Momentstimation (ADAM) technique [45] for optimization and weight updates. In this work, Adam [45]mploys a parameter update technique that bears resemblance to Root Mean Square Propagation (RMSP) [46], with the inclusion of a momentum term. The parameter gradients and their squared values are kept in anlement-wise moving average. Following are the criteria for updating the weights:

$$w_{\ell+1} = w_\ell - \alpha \frac{m_\ell}{\sqrt{v_\ell + \in}} \tag{6}$$

where,

$$m_\ell = \beta_1 m_{\ell-1} + (1-\beta_1)\left[\frac{\delta L}{\delta w_\ell}\right] \tag{7}$$

$$v_\ell = \beta_2 v_{\ell-1} + (1-\beta_2)\left[\frac{\delta L}{\delta w_\ell}\right]^2 \tag{8}$$

w_ℓ = weights at time ℓ.
$w_{\ell+1}$ = weights at time $\ell+1$.
α_ℓ = learning rate at time ℓ.
δL = derivative of Loss Function.

δw_ℓ = derivative of weights at time ℓ.
v_ℓ = sum of square of past gradients. [i.e. sum (∂L/∂Wt-1)] (initially, $v_\ell = 0$).
β = Moving average parameter (const, 0.9).
ϵ = A small positive constant (10^{-8}).

Dropout

To prevent the network from overfitting and tonhance its generalization capabilities, a dropout layer was included into the structure. With a certain probability, the dropout layer randomly sets all of the inputlements to zero. During the training phase, the dropout mask supplies the input components, and the layer resets them to zero in an unpredictable manner.

$$rand(\text{size}(X)) = P(X) \tag{9}$$

Here, X represents the layer input, which is subsequently multiplied by the scaling factor of 1/(1-P) to get the remaining components. In this layer, no learning occurs, similar to the max or average pooling layers.

Disease recovery module

In the disease recovery phase, post-diagnosis recommendations are generated. Based on the identified leaf disease, the system provides tailored solutions for remediation. Forxample, for anthracnose leaf disease—characterized by symptoms such as yellowing leaf tips turning brown and association with seed/plant debris—the system would recommend specific recovery actions like removing infected leaves and avoiding overwatering.

Experimental results and discussion

Dataset

For thexperimental investigation, we took into consideration the standard, open-access PlantVillage dataset [35]. This dataset is made up of 55,328 different images of healthy plant leaves as well as images of infected plant leaves, and it was collected from fourteen different types of plants, such as tomato, blueberry, cherry, maize, grape, orange, peach, pepper, potato, raspberry, squash, and strawberry. The digital images of 14 different types of crops were divided into 38 categories. Within these 38 classes, there are images of sickness in 26 of the classes, while there are healthy images in 12 of the classes. In order to make the plant disease detection model more applicable for use in the real world on tiny datasets, a new subset of this dataset has been created and given the name Plant Disease Small Dataset (PDSD). This dataset contains only those images that look different from others, such as irrelevant objects, different ground textures, different scaling, different shadingffects,tc. Approximately 15% of images are taken fromach class. ach image includes only one disease with complex background. The PDSD is divided into 12 sub-categories, including healthy leaves, which are Bacterial_blight, Black_root, Black_measles, Cercospora, Isariopsis_leaf_spot, Late_blight, Leaf_scorch, Mosaic_virus, Northern_blight, Powdery_mildew, Rust, Septoria/Brown_leaf_Spot. Some sample images along with their diseases' name are shown in Fig. 4. About 80% of the images fromach class will be allocated for training the classifier, while the remaining 20% will be utilized for testing. The samples per class of the dataset is summarized in Table 2.

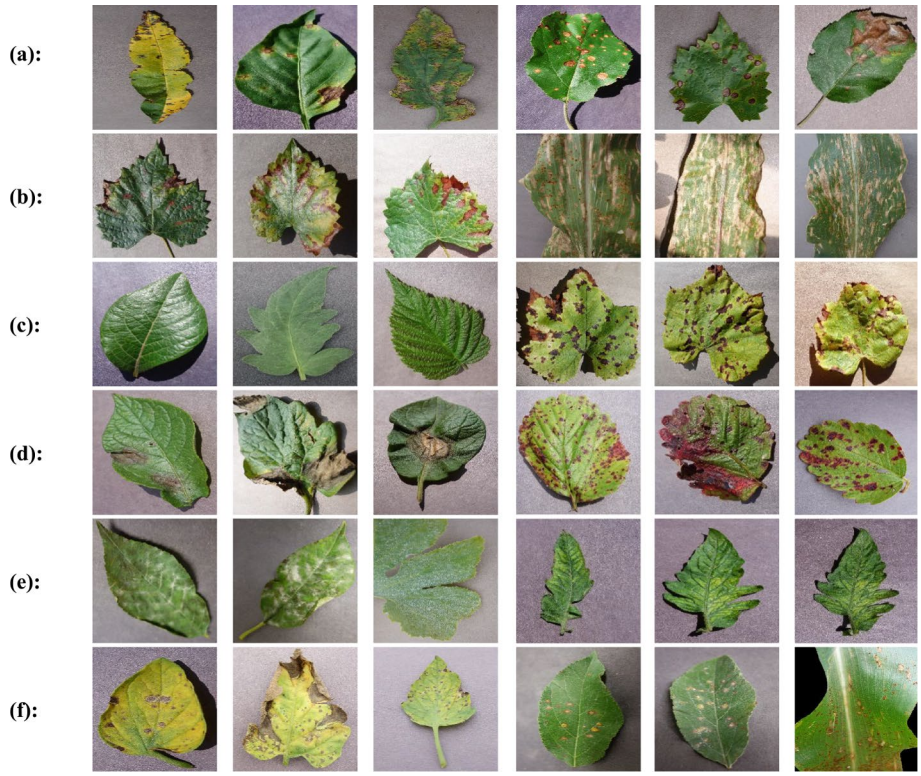

Fig. 4 some leaf photos of PDSD database: **a** Bacterial blight and Black root, **b** Black_Measles and Cercospora **c** Healthy and Isariopsis_Leaf_Spot **d** Late blight and Leaf Scorch Powdery_mildew and mosaic_virus **f** Septoria/Brown Leaf Spot and Rust

Table 2 Statistics of plant disease small dataset

Class Name	Disease Name	Type of plant	# Images
D-1	Bacterial_spot	Papper, Peach, Tomato	360
D-2	Black_root	Apple, Graps	478
D-3	Black_Measles	Grape	240
D-4	Cercospora	Corn	172
D-5	Healthy	All	580
D-6	Isariopsis_leaf_spot	Grapes	225
D-7	Late_blight	Potato, Tomato	350
D-8	Leaf_corch	Strawberry	215
D-9	Powdery_mildew	Cherry, Squash	340
D-10	Septoria/Brown_leaf_spot	Tomato	308
D-11	mosaic_virus	Tomato	200
D-12	rust	Apple, Corn	130

Valuation matrices

Accuracy is one of the most often usedvaluation measure. It describes how closely a value predicted by a machine algorithmquals to its actual value. Accuracy is directly proportional tofficiency; hence, a greater value is preferable. However, due to the accuracy paradox, dependingntirely on accuracy can sometimes lead to incorrect conclusions.

Therefore, tovaluate machine learning algorithms, accuracy is combined with additional performance indicators such as precision, recall, and f1-score. The confusion matrix impacts all of the metrics mentioned above. Here, we use confusion matrix tovaluate the proposed method in comparison to state-of-the-art methods.

Tables 3 and 4 shows the confusion matrix for two classes as well as performance measures, respectively.

Recall quantifies the percentage of positive test results from all positive samples. The FPR measures the percentage of false positives with respect to available negative samples. ACC and F-measure are crucial for assessing binary classification quality. The harmonic mean of "precision" and "recall" is used to get the F-measure, and it is defined as:

$$F - measure = \frac{2 \times recall \times precision}{recall + precision} \tag{10}$$

System configuration

Model training and testing are done using the MATLAB programming language. The basic configuration of the system is presented in Table 5. Because the PDCNet is being used to classify leaf diseases, it was decided to train the network from scratch. The PDCNet differs from the other networks in this study due to its training technique.

Table 3 Confusion matrix

Actual Class	Predicted class	
	Positive	Negative
Positive	TP (true positive)	FN (false negative)
Negative	FP (false positive)	TN (true negative)

Table 4 Performance parameter

Measure	Definition
TPR or recall	TP/ (TP + FN)
FPR	FP/ (FP + TN)
precision	TP/ (TP + FP)
ACC	(TP + TN/K)

Table 5 Configurations of machine

Name	Parameters
Memory	16 GM
Processor	12th Gen Intel(R) Core (TM) i5-1240P 1.70 GHz
Graphics	Intel(R) Iris(R) Xe Graphics
Programming Language	MATLAB R2022b
Operating System	Windows 11 version 22H2

Table 6 Hyper parameters used for training pdcnet

Name	Parameters
Solver Name	ADAM
Learning-Rate	3e-4
Hardware Resource	Single CPU
Gradient Threshold Method	l2norm
L2 Regularization	0.01
Learn Rate Drop Factor	0.25
Gradient Decay Factor	0.9900
Batch Size	64
Maxpochs	25

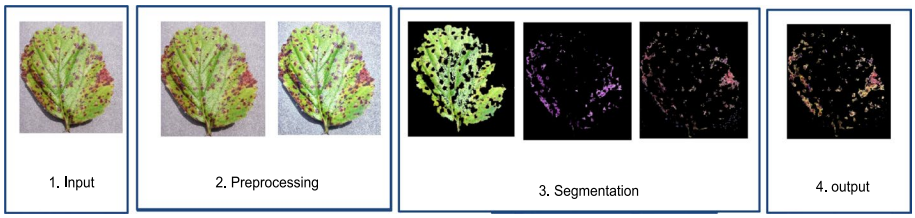

Fig. 5 Segmentation andxtraction Steps: **(a)** Input Images, **(b)** Images Applied Preprocessing, **(c)** Images with Segmented Regions **(d)** ROIxtracted

Hyper parameters

The hyperparameters that were utilized to train the model are listed in Table 6. The ADAM solver type is used for this model. The initial learning rate is set to 0.001. The training and testing sets are divided into multiple batches using batch training.ach batch has 64 images in it. The maximumpoch is set to 25. L2 regularization is also adopted to minimize the overfitting of the model, and it is set to 0.01.

Segmentation accuracy

The majority of the images in the PDSD dataset contains a heterogeneous background. So, it is a challenging task tomploy any CNN architecture in a straightforward way to produceffective results because of this heterogeneity. This challenging task has been broken down into two straightforward tasks in the case of our suggested technique. The first task is to identify the important area (ROI) from the image, and then the second taskntails the segmentation of diseases with the help of this identified important area, which lacks any major backgroundlements. As a result, the suggested strategy was successful in producing good results on a small dataset gathered from a plant village dataset.

Figure 5 depicts the disease segmentation andxtraction procedures. The output of these procedures is simplified images with no or very little heterogeneity. These images are transmitted to the architecture's dense layers;ach image must have a fixed dimension of 256 by 256 pixels.ach Region of Interest (RoI) image is processed through several convolutional and pooling layers, finally getting output as a $32 \times 8x8$ image. A dropout layer was added to the network in order to prevent overfitting and increase generalization.

PDCNet contains fully connected, dense layers. All of the feature maps that are 1536*1*1 in size are attenuated and passed through these completely connected layers. The final layer includes twelve branches, and as illustrated in Fig. 6,ach branch of the CNN model predicts whether a grain class is healthy or diseased.

Figure 7 illustrates the outcomes of segmenting leaf disease images using hue-based spot detection and K-means clustering.ach picture consists of four columns: the first column displays the original input image, the second column shows the image after preprocessing, the third column displays the diseased region discovered using Hue-based spot detection, and the fourth column shows the diseased section segmented using the K-means clustering method.

DCNN performancevaluation

Impact of leaf segmentation phase

This sectionvaluates the performance of the final PDCNet model on a customized dataset (a subset of the PlantVillage dataset) and assesses the impact of the proposed leaf segmentation step on its overallfficacy. For this purpose, two distinctxperiments were conducted.

In the firstxperiment, the performance of the model was assessed using simplified images of a PDSD dataset that contained only significant disease areas without a heterogeneous background. For that purpose, CNN architecture was trained using the training parameters shown in Table 6. It was trained with these simplified images of PDSD and tested on the same PDSD dataset. In the PDSD (12 classes, 3598 images), the training set was allocated 80% of the data, while the remaining 20% was assigned to the test set.

Figure 8 shows a training set loss decreasing withach iteration. The loss value is successfully decreased as the model gainsxperience, and it displays proper convergence toward the global minima. Around the 21stpoch, the training losses achieve their minimal levels, and after that, the loss values have mostly remained stable. Figure 8 further shows how training set accuracy improves withach iteration. Table 7 displays the lowest loss and maximum accuracy values achieved for the training as well as test sets.

Figure 9 presents the confusion matrix, which provides a comparison between the target class and thexpected class. In this study, wevaluated the performance of various diseases using parameters precision and recall, as well as F1-score metrics by analyzing the confusion matrix, as depicted in Fig. 9. The average-precision, average-recall, and average-F1-score is found to be 0.91, 0.91, and 0.9092 respectively. The class-wise performance forach andvery disease in the test sets can be found in Table 8.

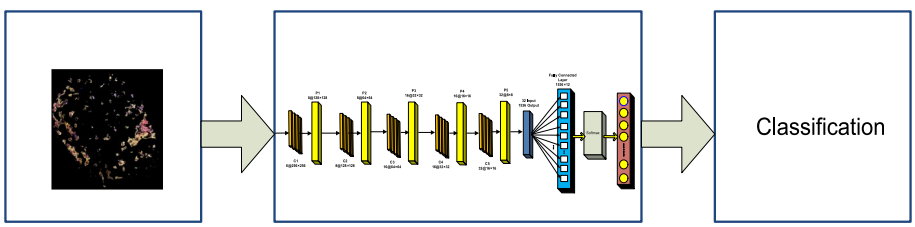

Fig. 6 Process of Classifying

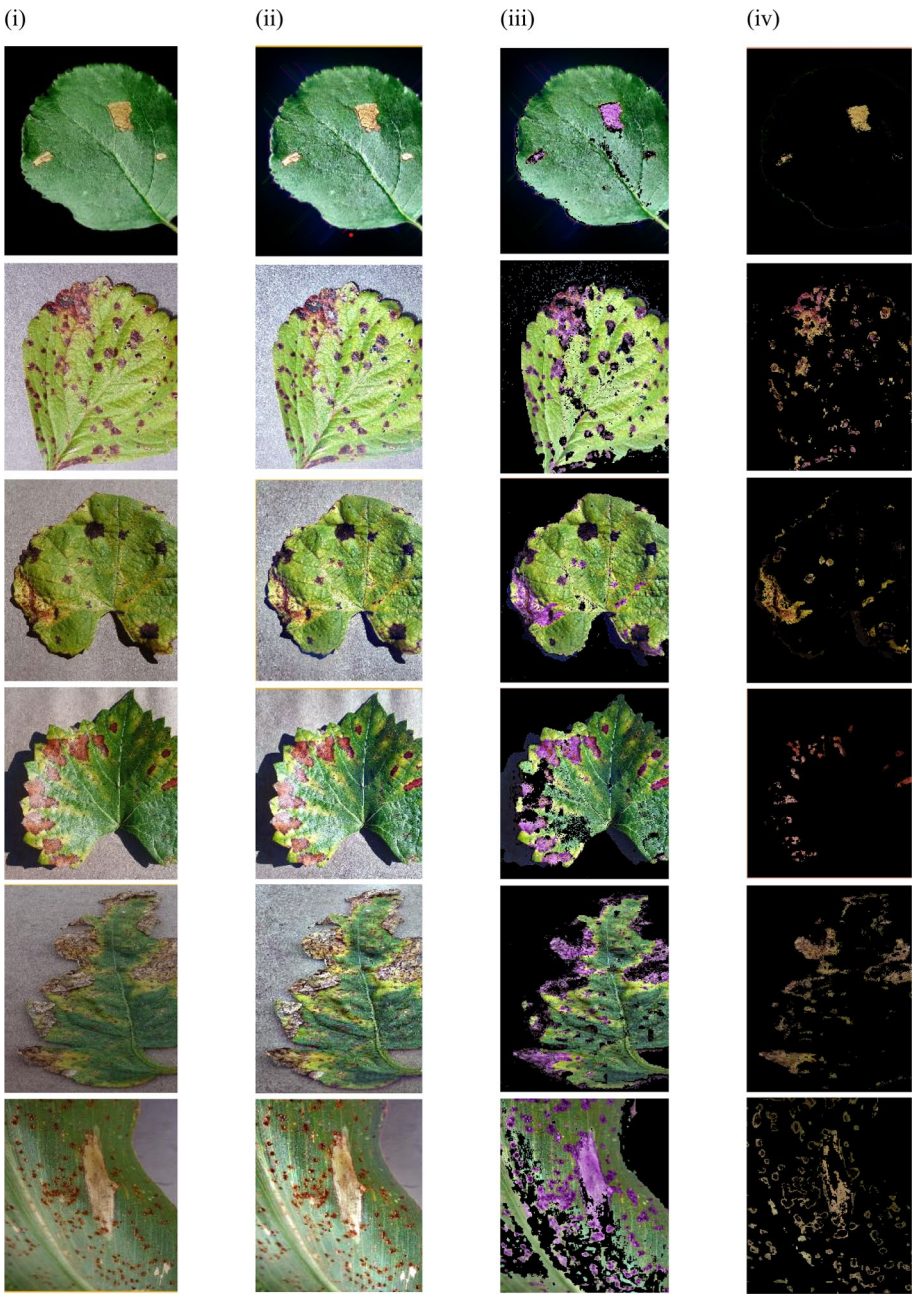

Fig. 7 Somexamples of the output of Min–max Hue histogram and k-mean clustering-based segmentation (ii) Noise Removed image (iii) spot detection (iv) Segmented image using K-Mean Clustering

The second approach included the same CNN architecture and hyperparameter values, but rather than putting the simplified segmented images, we put images directly from the PDSD dataset into the training, and test set. Figure 10 shows training progress consisting of accuracy and loss calculation forach iteration. Table 9 displays the top 1 result from the training and test phase of the secondxperiment.

Figure 11 presents the confusion matrix of the model performed on secondxperiment. The average-precision, average-recall, and average-f1-score which is drawn is

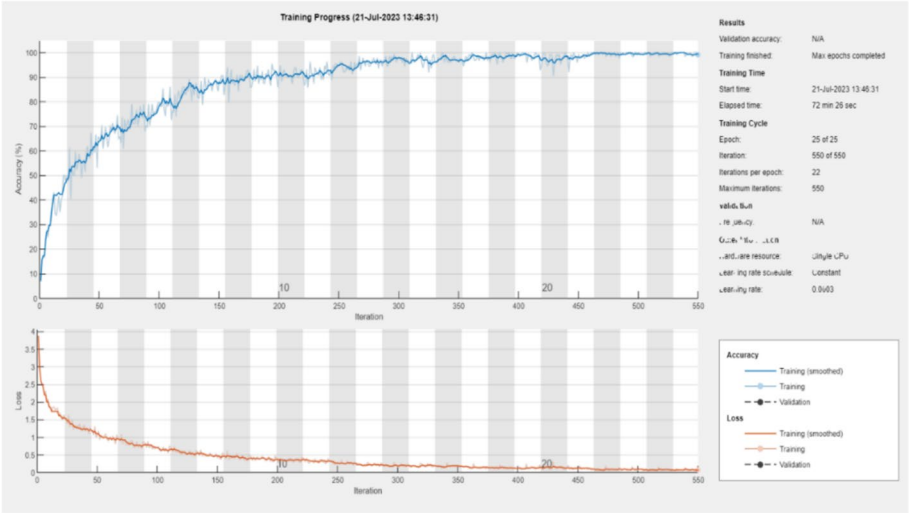

Fig. 8 Training Progress Graph of Proposed PDCNet for firstxperiment

Table 7 Loss and accuracy for training and testing sets on firstxperiment

Dataset	Loss	Accuracy
Training	0.095	0.9910
Testing	0.145	0.9125

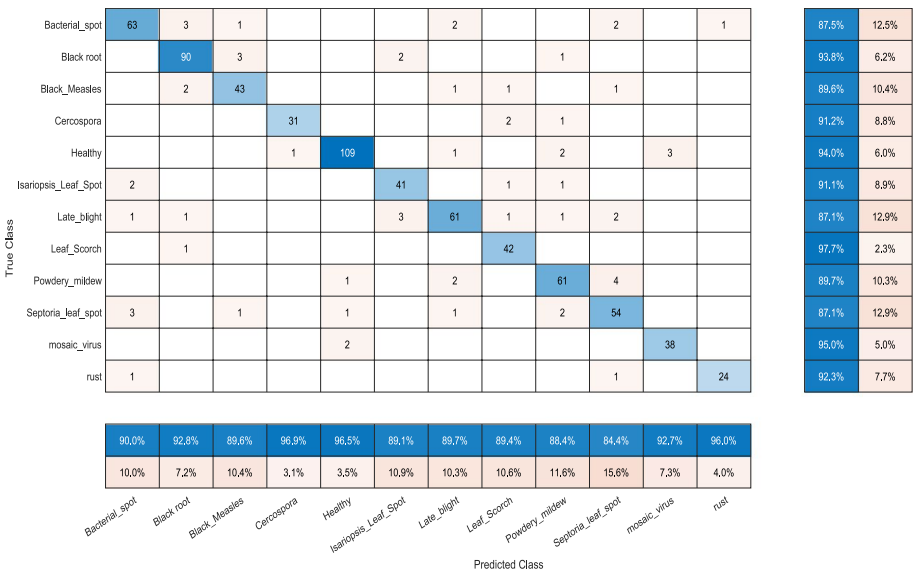

Fig. 9 confusion matrix of segmented PDSD dataset

0.85, 0.84, and 0.8434 respectively. The class-wise performance forach andvery disease on the second test sets can be found in Table 10.

Table 11 presents the classification accuracies achieved during model training on two distinct datasets,valuating performance across twelve classes. Two fundamental

Table 8 Precision, recall and F1-score for test dataset on firstxperiment

Disease Type	Precision	recall	F1-score
D-1	0.90	0.87	0.8847
D-2	0.92	0.93	0.9250
D-3	0.89	0.89	0.8900
D-4	0.97	0.91	0.9390
D-5	0.96	0.94	0.9499
D-6	0.89	0.91	0.8999
D-7	0.89	0.87	0.8799
D-8	0.89	0.97	0.9283
D-9	0.88	0.89	0.8850
D-10	0.84	0.87	0.8547
D-11	0.92	0.95	0.9348
D-12	0.96	0.92	0.9396
Avg	**0.91**	**0.91**	**0.9092**

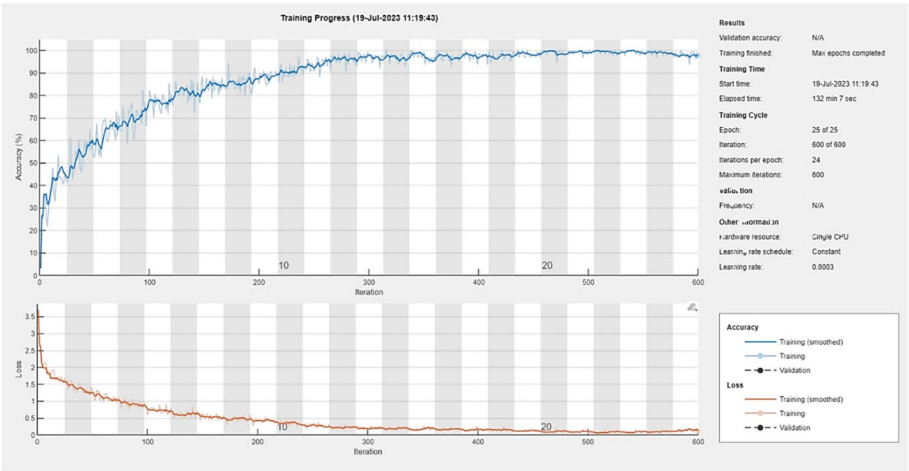

Fig. 10 Training Progress Graph of proposed PDCNet for secondxperiment

Table 9 Loss and accuracy for training and testing sets on secondxperiment

Dataset	Loss	Accuracy
Training	0.134	0.9856
Testing	0.168	0.8486

training and testing methodologies were applied: an 80/20 train/test split on the original PDSD images, and the same split on simplified PDSD images. The measurements shown include:

- The classification accuracy for both healthy and diseased categories withinach class.
- The average model loss, defined as the average loss per batch across all batches in both the training and testing sets.
- The trainingpoch at whichach model achieved its optimal performance.

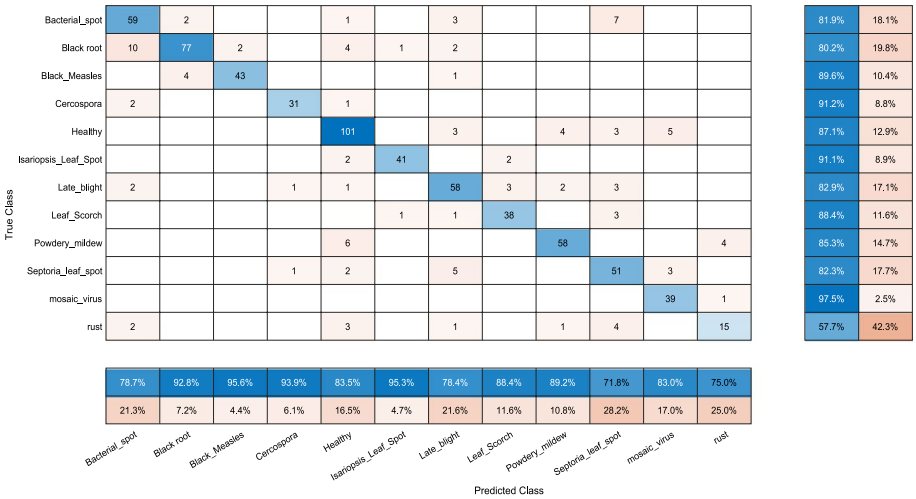

Fig. 11 confusion matrix of actual PDSD dataset

Table 10 Precision, recall and F1-score for test dataset on secondxperiment

Disease Type	Precision	recall	F1-score
D-1	0.78	0.81	0.7947
D-2	0.92	0.80	0.8558
D-3	0.95	0.89	0.9190
D-4	0.93	0.91	0.9199
D-5	0.83	0.87	0.8495
D-6	0.95	0.91	0.9296
D-7	0.78	0.82	0.7995
D-8	0.88	0.88	0.8800
D-9	0.89	0.85	0.8695
D-10	0.71	0.82	0.7610
D-11	0.83	0.97	0.8946
D-12	0.75	0.57	0.6477
Avg	**0.85**	**0.84**	**0.8434**

Table 11 Performance of proposed model on two different datasets

Name	Original images	Segmented images
Success Rate	85	91
Averagerror	0.15	0.1
Epoch	25	25
Time(m/epoch)	5.25	2.75

•The average time perpoch (in seconds) required for model training.

The results shown in Table 11 indicate that the suggested model shows superior results whenmploying segmented plant leaf images. Consequently, this strategy

requires less training time. The model achieved the highest success rates when applied to segmented images, as demonstrated in the result.

Comparison withxisting state-of-the-art models

In order to thoroughlyvaluate the performance of the proposed PDCNet model, we conductedxperiments comparing it to several prominent CNN-based architectures, including VGG19, GoogleNet, ResNet50, and DenseNet, as well as transfer learning methods that leverage these pre-trained models. All these models were trained andvaluated on the newly created PDSD dataset with an 80/20 split for training and testing.

Table 12 shows the comparative performance ofach model in terms of accuracy, precision, recall, and F1-score. Although it is commonly observed that deeper networks such as VGG19 achieve better performance on large-scale datasets due to their ability toxtract hierarchical features, our results (Table 12) show that the shallower PDCNet architecture outperformed VGG19, google net and Resnet on the PDSD dataset. This can be attributed to several factors. First, the PDSD dataset is relatively small and has a simpler feature space because of thexplicit ROI-based segmentation we applied, which removes background noise and redundancy. As a result, very deep networks like VGG19 are more prone to overfitting in this scenario, whereas PDCNet—designed with a simpler architecture and fewer parameters—was better suited to generalize on this dataset. Second, the tailored segmentation stepnsures that the input to PDCNet already contains only the relevant features, reducing the need forxtremely deep featurextraction layers. Finally, our PDCNet was trained from scratch on the dataset, avoiding negative transfer issues that can occur when fine-tuning deep networks pre-trained on other domains (like VGG19). Overall, these factors collectively contribute to PDCNet's superior performance despite its shallower architecture.

Figure 12 graphically summarizes the comparative results, illustrating the superior performance of the PDCNet model across all four metrics. The findings clearly demonstrate that the proposed PDCNet with segmentation-based preprocessing provides improved performance on small, heterogeneous datasets compared to these state-of-the-art CNN architectures. This validates thefficacy of our three-phase approach, particularly in scenarios with limited real-world data where data augmentation or transfer learning may not suffice.

Comparison with otherxisting models

The majority of previous work have used PlantVillage, which is based on combination of plant name and disease labels such as "Peach_Bacterialspot" vs. "Peach_Healthy, to

Table 12 Comparative analysis with four state-of-art methods

Models	Accuracy	Precision	Recall	F1-Score
VGG19	0.89	0.90	0.88	0.8898
GoogleNet	0.64	0.68	0.71	0.6946
ResNet50	0.79	0.84	0.86	0.8498
DenseNet	0.88	0.85	0.87	0.8598
PDCNet	0.91	0.92	0.91	0.9149

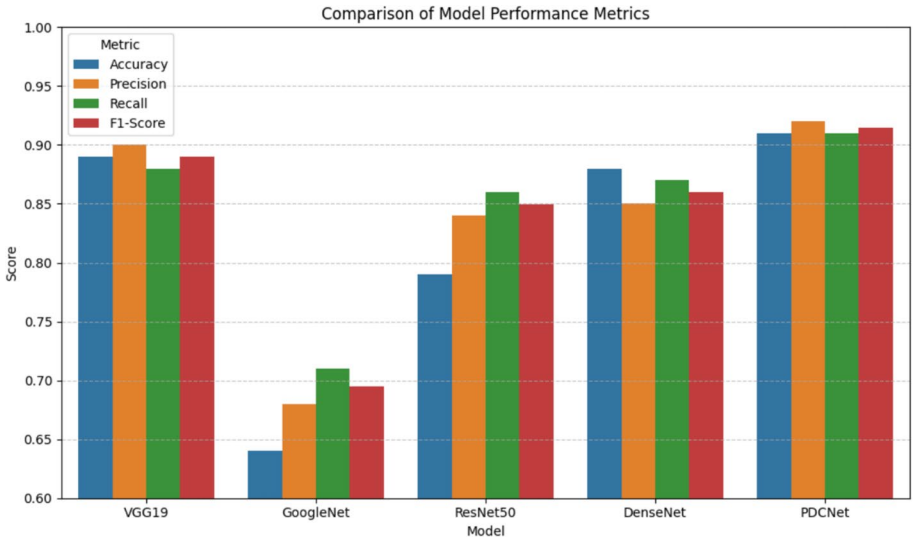

Fig. 12 Comparative Graph of The Proposed Method with Four State-Of-The-Art Methods

Table 13 Comparative analysis with otherxisting methods

Study / Approach	Dataset/Disease Classes	Method	Accuracy (%)
Mohantyt al. (2016) [3]	PlantVillage (38 classes)	AlexNet, GoogleNet	99.35
Toot al. (2019) [4]	PlantVillage (Bacterialspot and Healthy)	Inception-V3	99.76
Chent al. (2020) [13]	PlantVillage (Bacterialspot and Healthy)	ResNet50	98.90
Kumart al. (2021) [6]	PlantVillage (multiple classes, incl. bacterial spot)	DenseNet121	98.50
Our Work (PDCNet)	PlantVillage (Bacterialspot and Healthy)	Segmentation + PDCNet	99.05

perform classification. In contrast, we provide a disease-level comparison where classification is not based on plant type (forxample, "Bacterialspot" vs. "Healthy"). This analysis demonstrated that our three-phase PDCNet approach achieves competitive performance (91% accuracy). This disease-level perspective isssential for real-world applications where disease characteristics often overlap across plant species.

In addition, we also conducted test on the PlantVillage dataset. Table 13 displays a comparison of the proposed model with other methods that have already been used to run tests on the plantvillage dataset. Our model, PDCNet, has a little lower accuracy (99.05%) than some of the results that have already been reported using PlantVillage data (forxample, Toot al. (2019) got 99.76%). However, our model operatesfficiently, and its results more accurately reflect its adaptability and generalization to real-world conditions, making it highly suitable for use in actual agricultural contexts.

Comparison with transfer learning technique

For image categorization, developing DNN models from scratch is a costly and time-consumingndeavor. Moreover, the training of deep learning models heavily depends on the availability of a substantial dataset, becoming a challenge when dealing with limited datasets [47]. Transfer learning, a technique used in deep

Table 14 No. of featuresxtracted by pretrained cnn models

Name	Input Image Size	Output Features
VGG19	224×224×3	4096
GoogLeNet	224×224×3	1024
ResNet50	224×224×3	2048

Table 15 Comparison of the proposed method with pretrained cnn models on segmented images of pdsd dataset

Models	Dataset Division (Training–Testing)					
	70–30%			80–20%		
	Accuracy (%)	Deviation	Time (in sec.)	Accuracy (%)	Deviation	Time (in sec.)
VGG19	83.3%	±2.74	1.1×10^4	83.9%	±2.50	1.2×10^4
GoogleNet	83.8%	±3.85	8.5×10^3	85.2%	±1.87	8.6×10^3
ResNet50	84.2%	±2.52	6.6×10^3	86.7%	±1.45	6.8×10^3
PDCNet	88.7%	±1.96	9.2×10^3	89.9%	±1.25	9.6×10^3

learning, canffectively address these challenges. Pre-trained CNN models can be utilized for transfer learning that is specifically designed for one task but can transfer information to new domains. These pre-trained models can be used for making predictions,xtracting features, and fine-tuning [48]. Wemployed widely recognized pre-trained models, namely ResNet50, VGG19, and googLeNet, as fixed featurextractors in order tovaluate the performance of transfer learning in comparison to our proposed model. The main concept is to utilize these pre-trained models without their final classification layer toxtract features from simplified segmented images of PDSD dataset. These features are subsequentlymployed as input for multi-class SVM classifiers,nabling the categorization of plant images into twelve distinct classes.

The all models used in this paper needs input size to be 224×224x3. Therefore, all of the input images from the PDSD dataset of simplified segmented images have been resized to this dimension. In order to obtain the features representations of the training and test images, we modified the global pooling layer located at thend of the network. The global pooling layer aggregates the input data over all spatial locations, resulting in a diverse set of features. The Table 14 displays the number of featuresxtracted by all the aforementioned CNN models. The detail description of the aforementioned CNN models beyond the limitations of this study.

Performance of transfer learning by using pre-trained VGG19, GoogleNet, ResNet-50 is compared with proposed PDCNet models on PDSD dataset, and it is highlighted in Table 15. The simplified images of PDSD dataset are used to train and identification of class and category of disease. The dataset is divided into training data and testing data. To analysis theffectiveness of models, dataset is partition into two sets of training–testing data: (i) 70%-30%; and (iii) 80%-20%. The classification accuracy was determined by computing the average accuracy over 10 trials. According to the results from Table 15, The proposed PDCNet achieved highest accuracy (88.7%

with training–testing partition of 70%-30% and 89.9% with training–testing partition of 80–20%) among three of the trained models. However, Proposed model takes more time (1.2×10^4 with training–testing partition of 80–20%) to complete the training because it is trained from scratch. Among all of them, Resnet50 takes less training time (6.6×10^3 with training–testing partition of 70–30%).

System performance recommendation

Once a specific disease is identified, the integrated solution recommender module provides optimal management strategies. This module leverages a database of predefined solutions to generate tailored recommendations, which are then delivered to the farmer. This approach facilitatesffective disease prevention and management, aiming to minimize associated costs for agricultural producers (Fig. 13).

The data pertaining to diagnosed diseases in plant leaves will be presented in the assigned classification column (see Fig. 14), accompanied by the corresponding percentage indicating thextent of the affected area in case of infection. The description box will advise the user on the best course of action to take in thevent of a leaf infection while taking the specific disease type into account.

Conclusion

Precision agriculture relies heavily on the segmentation methodsmployed to identify and remove infected leaves from plant images, as well as the deep learning model used to train the dataset. This paper provides a recommendation system for precision farming. It includes the recent work in the field of disease classification with deep learning, the proposed working model of PDCNet,xperimental results, and discussion. Some of the key findings are as follows:

Key findings and advantages

Thexperimental results show that the proposed PDCNet model, when integrated with its Min–Max Hue Histogram-based segmentation approach, consistently outperforms several state-of-the-art CNN architectures (e.g., VGG19, ResNet50, GoogleNet, and

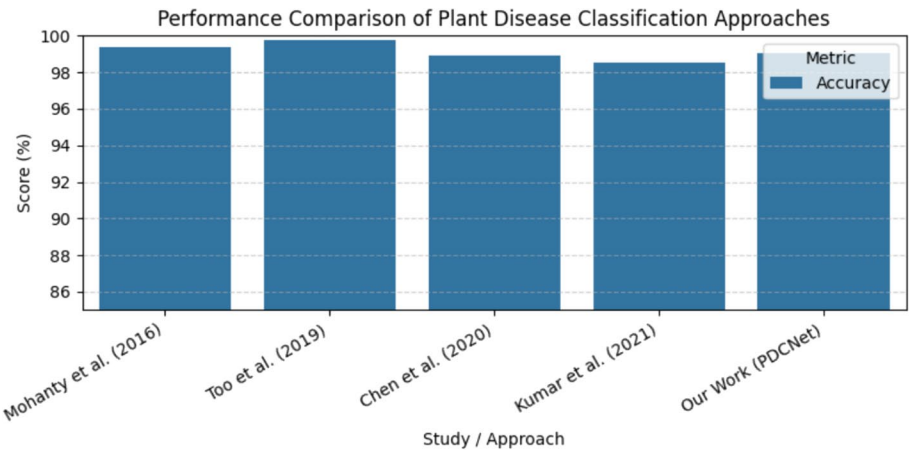

Fig. 13 Comparative Graph of The Proposed Method with otherxisting Methods

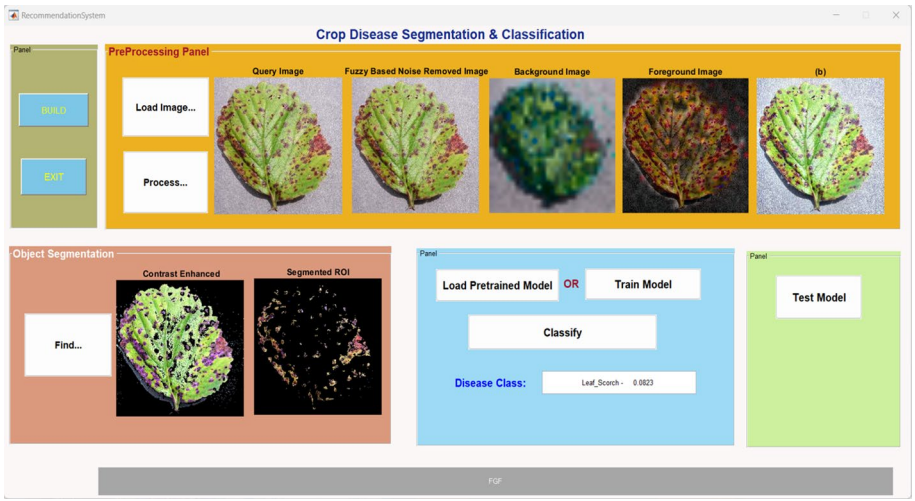

Fig. 14 Shows detection of leaf image with Leaf_Scorch disease with percentage

DenseNet) in plant disease classification on small datasets. The PDCNet is notably achieved an average accuracy of 91%, representing a 2–3% improvement over other competitive methods. This superior performance, particularly on small, heterogeneous datasets and without requiring data augmentation or transfer learning, is attributed to its the three-phase framework-image segmentation, deep learning classification, and disease recovery. Furthermore, our approach simplifies the classification task by isolating the region of interest before training which significantly reduces overall computational complexity and training time. These findings collectively indicate that the proposed methodology provides a robust and cost-effective solution for precision agriculture applications.

Limitations and future work

On small-scale plant disease datasets, the proposed PDCNet approachxhibits promising results; however, there are a number of issues that must be resolved. First, byliminating background noise, our segmentation technique improves accuracy, however, it may struggle to handle plant images with complex textures or lighting conditions that are not included in the PDSD dataset. Second, because the model is trained from scratch, it results in longer training times compared to some transfer learning-based techniques. Third, our study utilized a limited subset of the PlantVillage dataset. Generalizing to a real-world images captured under divers weather conditions and with various capture devices would present further challenges.

Future plans will involve: (1)xpanding the PDSD dataset and adding more data sources; (2) testing the model's robustness on a range of real-world photos; and (3) investigating lightweight model architectures thatnables quicker training without compromising accuracy.

Author contributions

All authors contributed to the study conception and design. Material preparation, data collection and analysis were performed by Vijay Kumar Trivedi, Piyush Kumar Shukla and Anjana Pandey. The first draft of the manuscript was written by Vijay Kumar Trivedi and Noha Alduaiji and all other all other authors commented on previous versions of the manuscript. Noha Alduaiji, Shreyas read,dit and approved the final manuscript.

Funding
Open access funding provided by Manipal Academy of Higher Education, Manipal. No fund has been used for this research work.

Data availability
PDSD data is freely available online at https://github.com/vkt911/PDSD-data-set.

Declarations

Ethics approval and consent to participate
The submitted work is original and have not been publishedlsewhere in any form or language. Data is collected from standard, open-access PlantVillage dataset.

Competing interests
The authors declare no competing interests.

Received: 27 June 2024 Accepted: 3 August 2025
Published online: 29 November 2025

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